

Name/UIN _____

An Aggie does not lie, cheat or steal or tolerate those who do!

Please answer the 5 multiple-choice questions and the written questions that follow. Good luck! This is a closed-book exam. You may use one *post-it* sticker note and a scientific calculator. Smartphones, tablets, smartwatches, laptops, and notebooks are not allowed.

1. (0.4pt) What is the main goal of the Metropolis–Hastings algorithm?
 - (a) To calculate exact probabilities in closed form.
 - (b) To minimize the variance of Monte Carlo estimates.
 - (c) To generate samples from a target probability distribution.
 - (d) To optimize a function using gradient descent.
 - (e) None of the above.
2. (0.4pt) In the classroom simulation of the Metropolis–Hastings algorithm, generating a proposal corresponds to?
 - (a) Choosing who to pass the ball to next.
 - (b) Calculating the acceptance ratio.
 - (c) Passing the ball to a randomly chosen student.
 - (d) Accepting or rejecting a move.
 - (e) None of the above.
3. (0.4pt) The acceptance probability in the Metropolis–Hastings algorithm is computed based on?
 - (a) The speed at which the ball is passed.
 - (b) The difference between current and proposed states using the target distribution.
 - (c) A random coin toss.
 - (d) The number of times the ball has been passed.
 - (e) None of the above.
4. (0.4pt) After computing the acceptance probability, what does the algorithm do next?
 - (a) Always accept the proposed pass.
 - (b) Decide whether to accept or reject the proposed move.
 - (c) Restart the algorithm from the beginning.
 - (d) Choose a new target distribution.
 - (e) None of the above.
5. (0.4pt) When the instruction says "try asymmetric proposals where the ball is passed right 70% of the time", what does this mean in algorithmic terms?
 - (a) The proposal distribution favors one direction more than another.
 - (b) The target distribution changes each time.
 - (c) The students pass the ball at random with equal probability.
 - (d) The acceptance probability must be exactly 0.7.
 - (e) None of the above.

6. a. (1pt) After having completed the classroom activity, explain in your own words the main purpose of the Metropolis–Hastings algorithm and how it relates to sampling from a probability distribution. Please explain how the algorithm works.

Answer:

- b. (1pt) Based on what you observed, describe how the class determined whether to accept or reject a proposed pass. What role did the acceptance probability play in this decision?

Answer:

- c. (1pt) What patterns did you notice when the acceptance probability was high compared to when it was low? How did this affect (or limited?) how the ball moved around the class?

Answer:

7. (1pt) Suppose that you have two dice. When you roll the first die, each face (1 through 6) is equally likely to appear. The other die is a loaded die that always lands on a 6. If we choose one of the dice at random and observe that the result is a 6, what is the probability that this die is the loaded one?

Answer:

8. (1pt) Suppose that you have three dice. Two are fair, each showing a 6 with probability $\frac{1}{6}$, and one is a loaded die that shows a 6 with probability 0.5. If you choose one die at random and observe that the result is a 6, what is the probability that this die is the loaded one?

Answer:

9. A smartphone company wants to estimate whether its new model s is *high-quality* or *low-quality* based on noisy customer ratings m from a small survey.

- $p(m \mid s)$: **Measurement distribution** — probability of observing a certain rating m given phone quality s .
- $p(s)$: **Prior belief** — how likely the phone is high- or low-quality before seeing any ratings.
- $p(s \mid m)$: **Posterior probability** — updated belief about phone quality after seeing the ratings.
- $p(m)$: **Evidence** — overall probability of observing the ratings.
- $p(r \mid s)$: **Response distribution** — probability that the company decides to *launch* or *redesign* given the true quality.

True or False? If false, please explain why.

- a. (1pt) Are the measurement distribution $p(m \mid s)$, prior $p(s)$, evidence $p(m)$, and posterior $p(s \mid m)$ correctly related by Bayes' rule as

$$p(s \mid m) = \frac{p(m \mid s) p(s)}{p(m)} ?$$

Answer:

- b. (1pt) The likelihood function is always the same as the measurement distribution.

Answer:

- c. (1pt) The quality level that maximizes the posterior probability is the same as the one that maximizes the likelihood.

Answer:

10. A small company wants to decide whether to launch a new product based on uncertain market demand. The company has prior information suggesting there is a 60% chance that market demand will be high. They plan to run a small pilot survey to update this belief before making a final decision.
- a. (1pt) Describe the decision problem and the hypotheses involved. State the hypotheses that represent the different possible market conditions. Briefly explain how the pilot survey will help the company make a better decision. *Note that: the main uncertainty concerns the level of market demand, which can be either **high** or **low**.*

Answer:

- b. (1pt) State how the prior probabilities and likelihood functions will be defined.

Answer:

- c. (1pt) Describe how the posterior probability will be computed and interpreted (*decision rule?*). Also, briefly identify some potential limitations or assumptions in your Bayesian approach.

Answer: